LUIS HENRICH-BANDIS

Solo Founder & Applied AI Engineer
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SUMMARY

Native-German technical founder. Built StudyPDF, an AI learning platform, to 140,000+ users and kept it profitable from day one. Decides what ships by measuring it first, with evaluation sets, LLM judges and controlled comparisons. Now studying the MSc Artificial Intelligence Applications and Innovation at Imperial College London.

EXPERIENCE

Founder & Sole Engineer · *StudyPDF, studypdf.net* **09/2024 – Present**

- **Grew StudyPDF to 140,000+ users as the only engineer**, profitable from day one. Students upload around 900 documents per day.
- **Built a gold evaluation set from 455 real user pages** for the vision pipeline that finds figures and tables. Annotated it by hand in Label Studio with a written labelling guide, including pages without figures to catch false positives.
- **Benchmarked 7 vision-language models on this set**. Rewriting one instruction block raised mIoU from 0.48 to 0.72, mainly by fixing boxes that captured too much of the page.
- **Built an evaluation harness for the RAG chat agent**: test cases checked by code and by an LLM judge, which blocks a release when quality drops below the baseline. Judges from three vendors rate answers blind, so no model grades its own output.
- **Built two-stage retrieval** (Pinecone search, then Cohere reranking) that also uses page images and cropped figures. Every citation is checked against the original page text before it is shown.
- **Routes each pipeline step to a model chosen by cost and quality**, across two providers with automatic failover. Found and fixed a JSON bug that was silently dropping 2.9% of vision responses.

AI Engineer, Working Student · *Capgemini, Cologne* **05/2026 – 07/2026**

- **Introduced Claude Code across a 35,000-file monorepo**: solved context-window problems, ran self-hosted models and compared code-exploration tools on speed, cost and quality.

AI Engineer Intern · *BMW Group, Munich* **10/2024 – 03/2025**

- **Shipped an internal RAG chat app** on a self-hosted LLM, so Virtual Commissioning engineers can ask questions about production planning documents.
- **Built an LLM tool** that generates state diagrams and technical documentation from product specifications.

EDUCATION

M.Sc. Artificial Intelligence Applications and Innovation · *Imperial College London* **09/2026 – Present**

Core modules: Machine Learning, Deep Learning, Responsible AI, AI Ventures. **Includes a supervised technical AI/ML placement** (ML engineering or AI research) — currently sourcing a host organisation.

M.Sc. Business Administration & Mechanical Engineering (coursework only) · *RWTH Aachen* **10/2025 – 09/2026**

ML coursework: Fundamentals of Machine Learning, Deep Learning in Economics, Machine Learning for Business Analytics.

B.Sc. Business Administration & Mechanical Engineering · *RWTH Aachen* **10/2021 – 09/2025**

Final GPA 1.8 ("good"), completed in standard period. Thesis 1.0 ("very good") on the business potential of a European energy data platform.

CS50x: Introduction to Computer Science · *Harvard University (online), 2023.*

SKILLS

Languages: German (native), English C1 (Duolingo English Test 135/160, July 2026).

Eval & measurement: gold sets and annotation (Label Studio), IoU/mIoU scoring, LLM-as-judge, blind judge panels, regression gates, A/B experiments.

LLM systems: multi-model routing and failover (OpenRouter, Azure AI Foundry), structured outputs, prompt caching, RAG with reranking, vision OCR and layout detection.

Engineering: TypeScript, Python, Bun/Node.js, Next.js, React, React Native, PostgreSQL/Drizzle, Redis + BullMQ, Pinecone, Docker, PostHog, Cloudflare R2.